

Unidad 4

DÍA

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FOLIO

Diferenciación en Integración numérica

4.1 Diferenciación numérica

1 $f(x) = x^2$, calcular $f'(2)$ con $h = 0.1$
Adelante, atrás y centrada

hacia adelante

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

$$f'(2) \approx \frac{f(2.1) - f(2)}{0.1}$$

$$f(2.1) = 2.1^2 = 4.41$$

$$f(2) = 2^2 = 4$$

$$f'(2) \approx \frac{4.41 - 4}{0.1}$$

$$f'(2) \approx 4.1$$

hacia atrás

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

$$f'(2) \approx \frac{f(2) - f(1.9)}{0.1}$$

$$f(1.9) = 1.9^2 = 3.61$$

$$f(2) \approx \frac{4 - 3.61}{0.1}$$

$$f'(2) \approx 3.9$$

Centrada

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$f'(2) \approx \frac{f(2.1) - f(1.9)}{2(0.1)}$$

$$f'(2) \approx \frac{4.41 - 3.61}{0.2}$$

$$f'(2) \approx 4.0$$

4.1, 3.9, 4.0

El valor exacto es:

$$f'(x) = 2x$$

$$f'(2) = 2(2) = 4$$

La centrada es la más precisa

2. $f(x) = \sin(x)$, $f'(\pi/4)$ con $h = 0.05$

Formula centrada

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$f'(\pi/4) \approx \frac{\sin(\pi/4 + 0.05) - \sin(\pi/4 - 0.05)}{2(0.05)}$$

$$f'(\pi/4) \approx \frac{\sin(0.8354) - \sin(0.7354)}{0.1}$$

$$\sin(0.8354) \approx 0.7416$$

$$\sin(0.7354) \approx 0.6710$$

$$f'(\pi/4) \approx \frac{0.7416 - 0.6710}{0.1}$$

$$f'(\pi/4) \approx 0.7068$$

$$f'(\pi/4) \approx 0.7071$$

El valor exacto sería:

$$f'(x) = \cos(x)$$

$$f'(\pi/4) = \cos(\pi/4) = 0.7071$$

3. $f(x) = e^x$, $f'(1)$ con $h = 0.1$

Comparar los 3 métodos

El valor exacto es:

$$f'(x) = e^x$$

$$f'(1) = e \approx 2.7183$$

hacia adelante

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

$$f'(1) \approx \frac{e^{1.1} - e^1}{0.1}$$

$$e^{1.1} \approx 3.0042$$

$$e^1 \approx 2.7183$$

$$f'(1) = \frac{3.0042 - 2.7183}{0.1}$$

$$f'(1) \approx 2.859$$

hacia atrás

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

$$f'(1) \approx \frac{e^1 - e^{0.9}}{0.1}$$

$$e^{0.9} \approx 2.4596$$

$$f'(1) \approx \frac{2.7183 - 2.4596}{0.1}$$

$$f'(1) \approx 2.587$$

Centrada

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$f'(1) \approx \frac{e^{1.1} - e^{0.9}}{0.2}$$

$$f'(1) \approx \frac{3.0042 - 2.4596}{0.2}$$

$$f'(1) \approx 2.723$$

La centrada es la más precisa

$$f'(1) \approx 2.723$$

4. $f(x) = \ln(x)$, $f'(2)$ con $h = 0.1$
Calcular error vs exacto

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$f'(2) \approx \frac{\ln(2.1) - \ln(1.9)}{2(0.1)}$$

$$\ln(2.1) \approx 0.7419$$

$$\ln(1.9) \approx 0.6419$$

$$f'(2) \approx \frac{0.7419 - 0.6419}{0.2}$$

$$f'(2) \approx \frac{0.1000}{0.2}$$

$$f'(2) \approx 0.500$$

Valor exacto

$$f(x) = \ln(x)$$

$$f'(x) = \frac{1}{x}$$

$$x = 2$$

$$f'(2) = \frac{1}{2} = 0.5$$

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$$\text{Error} = |\text{Valor exacto} - \text{Valor aproximado}|$$

$$\text{Error} = |0.5 - 0.500|$$

$$\text{Error} \approx 0.00$$

Aproximación : 0.500

Exacto : 0.5

Error : 0.00

5. $f(x) = x^3$, calcular $f''(1)$ con $h = 0.1$

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

$$f''(1) \approx \frac{f(1.1) - 2f(1) + f(0.9)}{(0.1)^2}$$

$$f(1.1) = (1.1)^3 = 1.331$$

$$f(1) = 1^3 = 1$$

$$f(0.9) = (0.9)^3 = 0.729$$

$$f''(1) \approx \frac{1.331 - 2(1) + 0.729}{0.01}$$

$$f''(1) \approx \frac{1.331 - 2 + 0.729}{0.01}$$

$$f''(1) \approx \frac{0.060}{0.01}$$

$$f''(1) \approx 6.0$$

Valor exacto

$$f(x) = x^3$$

$$f'(x) = 3x^2$$

$$f''(x) = 6x$$

$$f''(1) = 6(1) = 6$$

$$f''(1) \approx 6.0$$

Regla del Trapecio

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$$\int_0^1 x^2 dx \text{ con } n=4$$

$$\int_0^1 x^2 dx \quad n=4$$

Formula

$$\int_a^b f(x) dx \approx \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + f(x_n)]$$

$$h = \frac{b-a}{n}$$

$$h = \frac{1-0}{4} = 0.25$$

$$f(x) = x^2$$

x_0	$f(x_0)$
0	0
0.25	0.0625
0.5	0.25
0.75	0.5625
1	1

$$I \approx \frac{0.25}{2} [0 + 2(0.0625) + 2(0.25) + 2(0.5625) + 1]$$

$$I \approx 0.125 [0 + 0.125 + 0.5 + 1.125 + 1]$$

$$I \approx 0.125 (2.75)$$

$$I \approx 0.34375$$

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$$I \approx 0.3438$$

H = valor exacto
↓

$$\int_0^1 x^2 dx = \frac{1}{3} \approx 0.3333$$

Error aproximado

$$|0.3438 - 0.3333| = 0.0105$$

$$\int_0^{\pi} \sin(x) dx \text{ con } n=4$$

$$\int_0^{\pi} \sin(x) dx \quad n=4$$

Formula

$$\int_a^b f(x) dx \approx \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + f(x_n)]$$

$$h = \frac{\pi - 0}{4}$$

$$h = \frac{\pi}{4} \approx 0.7854$$

$$f(x) = \sin(x)$$

x_i	$f(x_i)$
0	0
$\pi/4$	0.7071
$\pi/2$	1
$3\pi/4$	0.7071
π	0

$$I \approx \frac{0.7854}{2} [0 + 2(0.7071) + 2(1) + 2(0.7071) + 0]$$

$$I \approx 0.3927 [1.4142 + 2 + 1.4142]$$

$$I \approx 0.3927 (4.8284)$$

$$I \approx 1.8961$$

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$$I \approx 1.8961$$

Valor exacto

$$\int_0^{\pi} \sin(x) dx = 2$$

Error aproximado

$$|2 - 1.8961| = 0.1039$$

$$\int_1^2 \frac{1}{x} dx \text{ con } n=4 \text{ calcular } \ln(2)$$

$$\int_1^2 \frac{1}{x} dx \quad n=4$$

$$h = \frac{2-1}{4} \quad h = 0.25$$

$$f(x) = \frac{1}{x}$$

x_i	$f(x_i)$
1.00	1.0000
1.25	0.8000
1.50	0.6667
1.75	0.5714
2.00	0.5000

$$I \approx \frac{0.25}{2} [1 + 2(0.8) + 2(0.6667) + 2(0.5714) + 0.5]$$

$$I \approx 0.125 [1 + 1.6 + 1.3334 + 1.1428 + 0.5]$$

$$I \approx 0.125 (5.5762)$$

$$\boxed{I \approx 0.6970}$$

Valor exacto

$$\int_1^2 \frac{1}{x} dx = \ln(2)$$

$$\ln(2) \approx 0.6931$$

Error aprox.

$$|0.6970 - 0.6931| = 0.0039$$

$$\int_0^1 e^x dx \text{ con } n=6$$

$$\int_0^1 e^x dx \quad n=6$$

$$h = \frac{1-0}{6} \quad h=0.1667$$

$$x_i = 0, 0.1667, 0.3333, 0.5, 0.6667, 0.8333, 1$$

$$f(x) = e^x$$

x_i	e^{x_i}
0	1.0000
0.1667	1.1814
0.3333	1.3956
0.5000	1.6487
0.6667	1.9477
0.8333	2.3010
1.0000	2.7183

$$I \approx \frac{0.1667}{2} [1 + 2(1.1814) + 2(1.3956) + 2(1.6487) + 2(1.9477) + 2(2.3010) + 2.7183]$$

$$I \approx 0.0833 [1 + 2.3628 + 2.7912 + 3.2974 + 3.8954 + 4.6020 + 2.7183]$$

$$I \approx 0.0833(20.6671)$$

$$\boxed{I \approx 1.7189}$$

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Valor exacto

$$\int_0^1 e^x dx = e - 1$$

$$e - 1 \approx 1.7183$$

Error aprox.

$$|1.7189 - 1.7183| = 0.0006$$

4.3 Regla de Simpson

10. $\int_0^1 x^2 dx$ con Simpson $1/3, n=4$

$$\int_0^1 x^2 dx \quad 1/3, n=4$$

Formula

$$I \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)]$$

$$h = \frac{1-0}{4} = 0.25$$

$$f(x) = x^2$$

x_i	$f(x_i)$
0	0
0.25	0.0625
0.50	0.2500
0.75	0.5625
1.00	1.0000

$$I \approx \frac{0.25}{3} [0 + 4(0.0625) + 2(0.25) + 4(0.5625) + 1]$$

$$I \approx 0.0833 [0.25 + 0.5 + 2.25 + 1]$$

$$I \approx 0.0833(4)$$

$$\boxed{I \approx 0.3333}$$

Es exacto porque:

$$\int_0^1 x^2 dx = \frac{1}{3} \approx 0.3333$$

$$\int_0^{\pi} \sin(x) dx$$

$$1/3 \quad n=4$$

$$I \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)]$$

$$h = \frac{\pi - 0}{4} \quad h = \frac{\pi}{4} \approx 0.7854$$

$$f(x) = \sin(x)$$

x_i	$f(x_i)$
0	0
$\pi/4$	0.7071
$\pi/2$	1
$3\pi/4$	0.7071
π	0

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$$I \approx \frac{0.7854}{3} [0 + 4(0.7071) + 2(1) + 4(0.7071) + 0]$$

$$I \approx 0.2618 [2.8284 + 2 + 2.8284]$$

$$I \approx 0.2618 (7.6668)$$

$$I \approx 2.0046$$

Valor exacto:

$$\int_0^{\pi} \sin(x) dx = 2$$

Error aprox.

$$|2.0046 - 2| = 0.0046$$

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$$\int_0^2 x^3 dx \quad n=6$$

$$I \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6)]$$

$$h = \frac{2-0}{6} \quad h = 0.3333$$

$$f(x) = x^3$$

x_i	$f(x_i)$
0	0
0.3333	0.0370
0.6667	0.2963
1.0000	1.0000
1.3333	2.3704
1.6667	4.6296
2.0000	8.0000

$$I \approx \frac{0.3333}{3} [0.4(0.0370) + 2(0.2963) + 4(1) + 2(2.3704) + 4.6296 + 8]$$

$$I \approx 0.1111 [0.148 + 0.5926 + 4 + 4.7408 + 18.5184 + 8]$$

$$I \approx 0.1111 (36)$$

$$I \approx 4.0$$

Valor exact

$$\int_0^2 x^3 dx = \frac{x^4}{4} \Big|_0^2$$

$$I = \frac{2^4}{4} = 4$$

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$$\int_1^3 dx \quad 1/3, n=4$$

$$h = \frac{3-1}{4} \quad h=0.5$$

$$f(x) = \frac{1}{x}$$

x_i	$f(x_i)$
1.0	1.0000
1.5	0.6667
2.0	0.5000
2.5	0.4000
3.0	0.3333

$$I \approx \frac{0.5}{3} [1.4(0.6667) + 2(0.5) + 4(0.4) + 0.3333]$$

$$I \approx 0.1667 [1 + 2.6668 + 1 + 1.6 + 0.3333]$$

$$I \approx 0.1667(6.6001)$$

$$\boxed{I \approx 1.1000}$$

$$\text{Valor exacto: } \int_1^3 \frac{1}{x} dx = \ln(3) \quad \ln(3) \approx 1.0986$$

Cuadratura Gaussiana

$$\int_{-1}^1 x^2 dx \text{ con 2 puntos Gauss}$$

$$[-1, 1]:$$

$$x_1 = -\frac{1}{\sqrt{3}}, \quad x_2 = \frac{1}{\sqrt{3}}$$

$$w_1 = w_2 = 1$$

$$f(x) = x^2$$

$$\int_{-1}^1 f(x) dx \approx w_1 f(x_1) + w_2 f(x_2)$$

$$I \approx \left(-\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2$$

$$I \approx \frac{1}{3} + \frac{1}{3}$$

$$I = \frac{2}{3} = 0.6667$$

$$\int_0^1 e^x dx \text{ transformando a } [-1, 1]$$

$$x = \frac{b-a}{2} t + \frac{a+b}{2}$$

$$a=0 \quad b=1$$

$$x = \frac{1}{2} t + \frac{1}{2}$$

$$dx = \frac{1}{2} dt$$

$$\int_0^1 e^x dx = \frac{1}{2} \int_{-1}^1 e^{\frac{1}{2}t + \frac{1}{2}} dt = \frac{1}{2} \left[e^{\frac{1}{2}t + \frac{1}{2}} \right]_{-1}^1$$

$$\frac{1 - \frac{1}{\sqrt{3}}}{2} \approx 0.2113$$

$$\frac{1 + \frac{1}{\sqrt{3}}}{2} \approx 0.7887$$

$$I \approx \frac{1}{2} (e^{0.2113} + e^{0.7887})$$

$$I \approx \frac{1}{2} (1.2353 + 2.2003)$$

$$I \approx 1.7178$$

$$e - 1 \approx 1.7183$$